

1. Problem Statement

Daily foot traffic can be very volatile while having a significant impact on retail stores and restaurants in urban high-traffic areas.

Having a rough idea of how many customers to expect based on historical data, taking weather conditions into account, can therefore help in better estimating occupancy rates based on foot traffic.

The system predicts the number of pedestrians on Bahnhofstrasse (Mitte) between Hauptbahnhof and Bürliplatz from the pedestrian's perspective, with an MAE of less than 150, which would represent a ~10% error in average over the past years traffic. As a reference, we use a naive baseline model: $\text{pedestrian_count_t} = \text{pedestrian_count_}(t-24)$. The main goal of the project is to focus rather on the pipeline implementation than on the model precision.

2. Originality & Motivation

The number of pedestrians on Bahnhofstrasse in Zurich is an economic factor that can directly impact the sales of nearby stores. On the one hand, employees walk along the street every day; on the other hand, there are customers with strong purchasing power. In addition, Bahnhofstrasse provides a direct connection from the main train station to Lake Zurich, which can attract many walk-in customers, especially when the weather is nice.

3. Data Sources & Features

Dataset Name	Source	Updatefrequency	Relevant Features
hystreet_fussgaengerfreq_since2022	Open Data Zürich - Stadt Zürich	Hourly (since 05.07.2022)	- Location Name - Temperature - Weatherlabel - Pedestrian_count (Label)
Meteoschweiz Automatic Station "Fluntern" (SMA)	Open Data (OGD) - MeteoSwiss	Hourly (last 10 years filtere do the project needs)	- Precipitation - Sunray - Snowfall

4. System Design

